Vectors Quiz, Version A

1.

Restate 215° in two other ways.

215° is between 180° and 270°: past west, toward south $215° - 180° = 35°$

Compass = **35° south of west**

$215° - 360° = -145°$

negative = **-145°**

same direction: $-145° = 35°$ south of west $= 55°$ west of south

Grading (2): 1 point for each correct form

2.

Restate 40° north of west in two other ways.

north of west is up and to the left: $180° - 40° = 140°$

0-360 = **140°**

$140° - 360° = -220°$

negative = **-220°**

same direction: $140° = -220° = 50°$ west of north

Grading (2): 1 point for each correct form

3.

Restate −70° in two other ways.

$-70° + 360° = 290°$

0-360 = **290°**

290° is 70° short of 360°: below east

Compass = **70° south of east**

same direction: $290° = 20°$ east of south $= 70°$ south of east

Grading (2): 1 point for each correct form

4.

Pac-Man travels 45 m north and then 28 m west through the maze. Find his resultant displacement: magnitude and direction.

$x = -28$ m $y = +45$ m $R = \sqrt{((28)^2 + (45)^2)} = \sqrt{2809} = 53$ m

$R = \boxed{53 \text{ m}}$

$\tan^{-1}\frac{45}{28} = 58.1^\circ$ north of west reference angle

~~$\theta = 58.1^\circ$~~ that is measured from west; the vector points up and left

$\theta = 180^\circ - 58.1^\circ = 121.9^\circ$

$\theta = \boxed{122^\circ}$

same direction: $-238^\circ = 32^\circ$ west of north $= 58^\circ$ north of west

Grading (5): sketch 1 • R setup 1 • R and unit 1 • tan setup 1 • direction 1

5.

Sam Wilson flies the Falcon at 240 km/h at 40° north of east. A wind blowing toward 30° south of east at 65 km/h also acts on him. Find his resultant velocity: magnitude and direction.

Convert to the 0-360° scale

P: 40° north of east $= 40^\circ$ W: 30° south of east $= 360^\circ - 30^\circ = 330^\circ$

$P_x = 240 \cos 40^\circ = 183.85$ km/h $P_y = 240 \sin 40^\circ = 154.27$ km/h

$W_x = 65 \cos 330^\circ = 56.29$ km/h $W_y = 65 \sin 330^\circ = -32.50$ km/h

$R_x = 240.14$ km/h $R_y = 121.77$ km/h

$R = \sqrt{((240.14)^2 + (121.77)^2)} = 269.25$ km/h

$R = \boxed{270 \text{ km/h}}$

$\tan^{-1}\frac{121.77}{240.14} = 26.9^\circ$ quadrant I, no adjustment

$\theta = \boxed{27^\circ}$

same direction: $-333^\circ = 27^\circ$ north of east $= 63^\circ$ east of north

Grading (5): components 2 • add components 1 • R 1 • direction 1